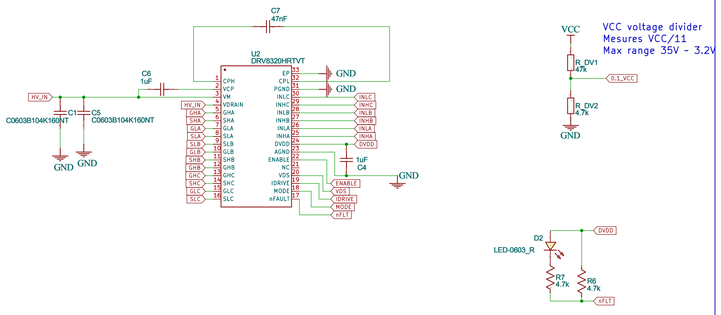
DRV8320H Gate driver circuit

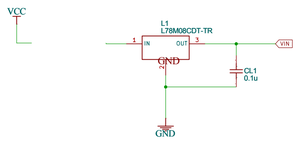


INA240 Current Sense Circuit

INA240A1: gain 20
Shunt resistor: 4mOhm/2 (parallel)
Cumulative Gain: 0.002*20 = 0.04 V/A
Range: 3.3/2/0.04 = +-41A



L7808 Circuit
+ Enable/Disable config



VDS config



VDS
voltage between SHx and SLx
limiting the voltage to limit the current
max current 50Amps (t_OCP= 4us impulse)
drain-source resistance 4mOhm
VDS = 50A*0.004Ohm = 0.2V

configuration
VDS = Tied to AGND 0.06V
VDS = 18 kΩ ± 5% tied to AGND 0.13V
VDS = 75 kΩ ± 5% tied to AGND 0.26V (BEST)
VDS = Hi-Z 0.6V
VDS = 75 kΩ ± 5% tied to DVDD 1.13VV
VDS = 18 kΩ ± 5% tied to DVDD 1.8V

IDRIVE config



IDRIVE
Setting the target gate current
directly determining the mosfet rise time
our mosfet gate-source charge Q = 3nC
rise time = Q/IDRIVE (pp 66)
we want the rise time of 50ns
IDRIVE = Q/50ns = 0.06A

configuration
IDRIVE = Tied to AGND 10mA
IDRIVE = 18 kΩ ± 5% tied to AGND 30mA
IDRIVE = 75 kΩ ± 5% tied to AGND 60mA (BEST)
IDRIVE = Hi-Z 120mA
IDRIVE = 75 kΩ ± 5% tied to DVDD 260mA
IDRIVE = 18 kΩ ± 5% tied to DVDD 570mA
IDRIVE = Tied to DVDD 1000mA

MODE config

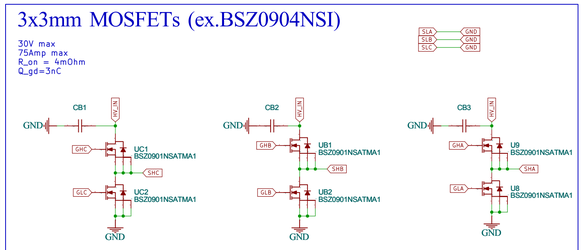


MODE
Setting the 3xPWM mode
Option for 6xPWM as well
pp 31

configuration
MODE = 00b = Tied to AGND 6xPWM
MODE = 01b= 47 kΩ ± 5% tied to AGND 3xPWM
...

3x3mm MOSFETs (ex.BSZ0904NSI)

30V max
75Amp max
R_on = 4mOhm
Q_gd=3nC



DRV8320H pinout Config



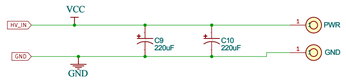
Encoder/Hall pinout config



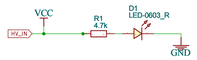
Current Sense Pin Config



Capacitors



LED Circuit



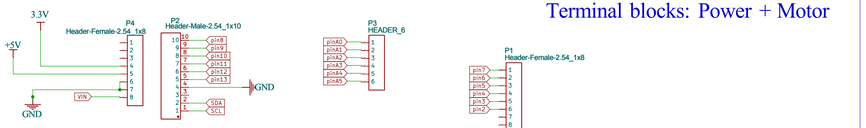
5/3.3V VDD selector



Enable/Disable Pullups Config



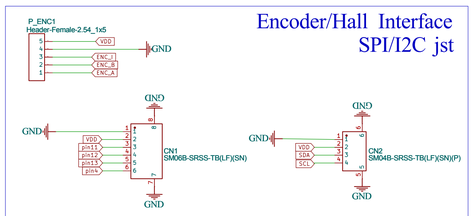
Arduino Header R3
Encoder/Hall Interface
Terminal blocks: Power + Motor



Motor output terminal



Encoder/Hall Interface
SPI/I2C jst



TITLE: SimpleFOC DriveShield	REV: 1.8
EasyEDA Company: SimpleFOC	Sheet: 1/1
Date: 2026-06 Drawn By: the.skuric	